

ROMAN DIMAKOV

Curriculum Vitae – Jan 2026

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RESEARCH INTERESTS

Economics of Science, Economics of Innovation, Social Network Analysis, Natural Language Processing

EDUCATION

CERGE-EI , Prague, Czech Republic Ph.D. in Economics	2020 - present
Bocconi University , Milan, Italy Visiting Scholar, hosted by Prof. Mara P. Squicciarini	Spring 2025
Lomonosov Moscow State University MSc Economics	2015-2017
Lomonosov Moscow State University BA Economics	2011-2015

WORK IN PROGRESS

”Diffusion of Knowledge and Productivity Effects in the Social Networks of Scientists: Evidence from the Discovery of Quantum Mechanics”

Abstract. This paper aims to provide causal evidence on the impact of a scientific breakthrough on scientists’ productivity and diffusion of knowledge by examining one of the major scientific discoveries in the twentieth-century physics – the discovery of quantum mechanics. I collect a dataset of historical scientific correspondence covering more than 5,000 active physicists between 1900 and 1952. Using this dataset, I map correspondence networks among physicists and approximate the flow of knowledge among them through the intensity of their letter exchanges. I complement the correspondence dataset with scientists’ publication and citation data from the Web of Science. I exploit pre-discovery correspondence links to define scientists’ exposure to the ideas of quantum mechanics based on their proximity to the theory’s key founders. I then study individual productivity outcomes, such as the number of published papers and their quality (measured by forward citations from the Web of Science database). To estimate the causal effect of exposure to a scientific revolution, I employ a difference-in-differences framework supported by inverse probability weighting based on individual-level network characteristics. Preliminary results suggest that physicists who had established correspondence with the founders of quantum mechanics prior to the discovery produced more papers of higher quality – measured by forward citations within five and twenty years after publication – relative to comparable scientists without such connections. This study contributes to the growing literature on the diffusion of ideas, the role of social networks in science, and the determinants of researchers’ productivity.

Key words: Quantum mechanics, social networks, productivity, science

”Sources of Upper-Tail Human Capital: Evidence from the Prussian Education Reforms in the Early 19th Century” (with Konstantin Bakharev and Vladimir Shchukin)

Abstract. How can a nation become a global leader in science and technology? Investment in education can boost human capital, which is crucial for economic development. This project examines whether large-scale education reforms, exemplified by the Prussian education reforms of the early 19th century, can cultivate upper-tail human capital (advanced skills and knowledge). After its defeat by Napoleon, Prussia initiated nation-wide reforms to modernize the state. A core component of these reforms were education reforms that focused on natural sciences and technology. We exploit natural variation in cities' exposure to the education reforms, comparing Prussian cities affected by the reforms with cities in other German-speaking states that were unaffected. We approximate upper-tail human capital by analyzing the births and deaths of prominent scientists at the city level. Using a Difference-in-Differences empirical framework, we estimate the effects of the reforms on upper-tail human capital formation. Our project contributes to two areas of economic literature: the origins of (upper-tail) human capital and the study of notable individuals. To the best of our knowledge, we are the first to systematically explore the causal impact of national education reforms on upper-tail human capital formation, focusing on Prussia's experience. This study sheds light on the origins of the highly skilled population that contributed to Germany's scientific and industrial success in the late 19th century.

Key words: Prussia, upper-tail human capital, reforms, education

RESEARCH AND RELEVANT EXPERIENCE

Research Assistant to Paolo Zacchia, CERGE-EI	Spring & Summer 2023
Research Assistant to Christian Ochsner, CERGE-EI	Fall 2021
Leading Business Analyst , Fleetcor Technologies, Inc., Moscow	2019-2020
Risk Manager , Bank Otkritie Financial Corporation, Moscow	2018-2019
Risk Analyst , Alfa Bank, Moscow	2018

TEACHING

TA for ML for Social Scientists , CERGE-EI	Fall 2026
TA for Statistics: Foundations of Data Science , CERGE-EI	Fall 2025
TA for Statistics: Foundations of Data Science , CERGE-EI	Fall 2024

SUMMER SCHOOLS

Economics of Ideas, Science and Innovation – The Institute for Progress Short Ph.D. course (online)	2024
Networks in Micro- & Macroeconomics – Tinbergen Institute Summer school	2023
Unstructured Data in Economics – CEMFI Summer school (online)	2023

OTHER PROFESSIONAL ACTIVITIES

Conference of the European Association of Labour Economists (EALA) Co-organizer, Prague, Czech Republic	2023
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GRANTS, AWARDS AND SCHOLARSHIPS

Research grant, the Charles University Grant Agency (GA UK) <i>~ €10,000 a year</i>	2025-2026
Doctoral Fellowship, CERGE-EI	2020 - present
Full Scholarship, Lomonosov MSU	2015-2017
Full Scholarship, Lomonosov MSU	2011-2015

PRESENTATIONS

2026:

CERGE-EI Applied Workshop (Mar 2026)

2025:

International Scientific Conference on Economic Policy (ISCEP), organised by ERUNI, Czech Republic (Sep 2025); Bocconi University seminar (Mar 2025); Brown Bag Seminar, CERGE-EI (Feb 2025)

2024:

Brown Bag Seminar, CERGE-EI (May 2024)

LANGUAGES & TECHNICAL SKILLS

Languages: English, Russian (native speaker)

Technical skills: Python, Stata, R, Gephi, Ollama, SQL, MS Excel

PERSONAL INFORMATION

Nationality: Belarus, born on April 9th, 1994

Currently resident in the Czech Republic